



NLP ESSENTIALS - QUICK GUIDE



Phrases, Parsing & Embeddings Made Simple

1 WHAT IS PHRASE?

A phrase is a group of words that work together as a unit but **do not contain a subject and predicate.**



Examples:

- in the garden
- playing tennis
- very quickly

2 TYPES OF PHRASE

Noun Phrase (NP)



Acts like a noun

Example:
the red car

Verb Phrase (VP)



Acts like a verb

Example:
is running

Adjective Phrase (AdjP)



Acts like an adjective

Example:
very beautiful

Adverb Phrase (AdvP)



Acts like an adverb

Example:
quite silently

Prepositional Phrase (PP)



Starts with a preposition

Example:
in the garden

3 WHAT IS PARSING?

Parsing is the process of analyzing a sentence to understand its **grammatical structure.**

Example Sentence:

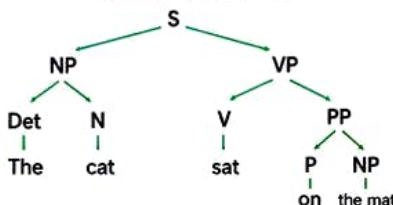
The cat sat on the mat.



4 TYPES OF PARSING

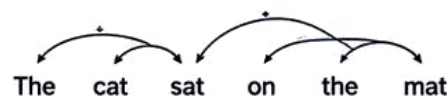
Constituency Parsing
(Structure based)

Shows how words group together in a hierarchy (tree).



Dependency Parsing
(Relationship based)

Shows how words are related to each other.



5 WHAT IS WORD EMBEDDING?

Word embedding is a way to represent words as **real-valued vectors** in a continuous vector space, so that words with **similar meaning** are closer in that space.



6 EMBEDDING SCHEME TYPES

One-Hot Encoding

1 0 0 0 0

Simple but high dimensional and sparse.

Count-Based
(BoW, TF-IDF)



Based on word frequency or co-occurrence.

Predictive
(Word2Vec)



Predicts context words from target word.

GloVe



Uses global word co-occurrence statistics.

FastText



Considers subword information.

7 WHAT IS CONTEXTUAL EMBEDDING?

Contextual embedding generates word representations based on the context in which a word appears. So the same word can have **different vectors in different sentences.**

I deposited money in the **bank**. → Financial meaning

We sat on the **bank** of the river. → River edge meaning

8 CONTEXTUAL EMBEDDING SCHEME TYPES

ELMo



Deep contextual embeddings using bi-LSTM.

BERT



Uses Transformer (encoder) for deep context.

GPT



Uses Transformer (decoder) for generative tasks.

RoBERTa



Improved BERT with more data & optimization.

XLNet



Captures context in both directions with permutations.

SUMMARY FLOW



These concepts are the building blocks of **Natural Language Processing (NLP)**!